WINDOW SERVER 2025

Active Directory Administration Lab

Project Documentation

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Platform: Oracle VirtualBox

Project type: Personal learning lab / IT portfolio

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Document Control

Project name	Windows Server 2025 – Active Directory Administration Lab
Author	Musa CHanzi Sichone
Environment	Oracle VirtualBox virtual machine
Status	In progress update after completing and testing each task
Repository	Add your GitHub repository URL after publishing

1. Project Overview

This project documents the setup and administration of a Windows Server 2025 virtual lab. The intended outcome is to practise core system-administration tasks, including network configuration, Active Directory Domain Services (AD DS), DNS, domain user management, Group Policy, and client-domain joining. The project is for learning and portfolio demonstration in an isolated test environment.

Important: this document is a working template. Mark a task as completed only after you have performed it and captured evidence from your own virtual machine.

2. Objectives

- Install and prepare Windows Server 2025 in Oracle VirtualBox.
- Configure and verify a static IPv4 address for the server.
- Install AD DS and configure a test domain.
- Configure DNS and verify name resolution.
- Create organizational units, test users, and security groups.
- Join a compatible Windows client virtual machine to the domain.
- Create and test a basic Group Policy setting.
- Record configuration steps, test results, screenshots, and lessons learned.

3. Scope and Lab Design

The lab simulates a small organization's internal Windows network. The server is intended to provide centralized identity and authentication services. A separate Windows client VM is recommended for testing domain joining and policy application. Do not expose a test domain controller directly to the public internet.

Component	Planned role	Actual value
Host computer	Runs VirtualBox	Window server 2025
Server VM	Windows Server 2025	RAM 8GB,HDD 50GB
Client VM	Windows client	Windows 11
Virtual network	Connects server and client	NAT
Domain name	DC01	IPv4: 10.10.2.15
Server IPv4	Static address	IPv4: 10.10.2.15
DNS server	Domain controller	127.0.0.1

4. Requirements

- A computer capable of running Oracle VirtualBox and the selected virtual machines.
- Windows Server 2025 installation media obtained from a legitimate Microsoft source.
- Adequate memory and disk space allocated to the VMs.
- A Windows client edition that supports joining an Active Directory domain (verify edition requirements).
- Administrator access to the lab VMs.
- A folder for screenshots, notes, scripts, and test results.

5. Implementation Procedure

5.1 Install Windows Server 2025

- Create a new virtual machine in VirtualBox and attach the Windows Server 2025 ISO.
- Install Windows Server using the selected Desktop Experience option if a graphical interface is required.
- Set a strong local Administrator password and sign in.
- Install any appropriate VirtualBox guest additions only if supported and needed.
- Rename the server to a clear lab name, for example DC01, then restart if prompted.
- Record the VM resources and capture a screenshot of Server Manager or the System page.

5.2 Configure a Static IP Address

Choose an IP plan that matches the VirtualBox network you created. Do not copy the example values below without checking your own network. The server and client must be able to communicate on the same reachable lab network.

Setting	**Example**
IPv4 address	10.10.2.15
Subnet mask	255.255.255.0
Default gateway	10.10.2.2
Preferred DNS	127.0.0.1

5.3 Install Active Directory Domain Services

- Open Server Manager and select Add Roles and Features.
- Choose Role-based or feature-based installation and select the local server.
- Select Active Directory Domain Services and accept required management tools.
- Complete the wizard and select the notification flag to promote the server, if appropriate.
- For a new lab forest, choose Add a new forest and enter a domain name you control for the lab (for example, a reserved internal lab name).
- Set the Directory Services Restore Mode password and store it securely; do not put it in GitHub or this document.
- Review prerequisites, install, and allow the server to restart.
- Sign in using the domain administrator credentials and verify that AD DS tools are available.

Record the actual domain name: [Fill in after configuration].

Evidence to capture: AD DS role installed, domain details, and Active Directory Users and Computers.

Status: ☐ Not started ☐ In progress ☐ Completed and tested

5.4 Verify DNS

- Confirm that the DNS Server role is installed if it was selected during domain-controller promotion.
- On the server, run: ipconfig /all
- Run: nslookup <your-domain-name>
- Run: dcdiag /test:dns (run from an elevated command prompt if the tools are available).
- Record the result and investigate any errors before proceeding.

Evidence to capture: successful DNS lookup and any relevant diagnostic output.

Status: ☐ Not started ☐ In progress ☐ Completed and tested

5.5 Create Organizational Units, Users, and Groups

- Open Active Directory Users and Computers.
- Create a lab OU such as LabUsers and another such as LabComputers if useful.
- Create a test user with a non-sensitive lab-only account name.
- Create a security group and add the test user to it.
- Verify that the objects appear in the intended OU and group membership is correct.
- Use test accounts only; never upload real passwords or personal data.

Evidence to capture: OU tree, test user properties (without secrets), and group membership.

Status: ☐ Not started ☐ In progress ☐ Completed and tested

5.6 Join a Windows Client to the Domain

- Create or prepare a separate Windows client VM whose edition supports domain joining.
- Connect it to the same VirtualBox network as the server and confirm basic connectivity.
- Set the client's preferred DNS server to the domain controller's IP address.
- Test name resolution and connectivity before attempting domain joining.
- Open the system's domain/workgroup settings and choose to join the domain.
- Enter the lab domain name and use an authorized domain account when prompted.
- Restart the client, sign in with the test domain user, and verify the computer object in Active Directory.

Evidence to capture: client domain membership and its computer object in Active Directory.

Status: ☐ Not started ☐ In progress ☐ Completed and tested

5.7 Configure and Test Group Policy

- Open Group Policy Management on the domain controller.
- Create a clearly named test GPO and link it to the intended lab OU or domain as appropriate.
- Choose a low-risk setting suitable for a test environment; document the setting and scope before applying it.

- On the client, run: gpupdate /force
- Use: gpresult /r to review applied Group Policy objects.
- Verify the expected behavior and document the result. Avoid policies that could lock you out of the lab.

Evidence to capture: GPO settings and gpresult /r output.

Status: ☐ Not started ☐ In progress ☐ Completed and tested

Test	**Expected result**	**Actual result / evidence**	**Status**
Server has planned static IP	Address matches network plan	[Add result]	Not tested
Domain name resolves through DNS	nslookup returns appropriate result	[Add result]	Not tested
AD user and group created	Objects visible in AD tools	[Add result]	Not tested
Client joins domain	Client reports domain membership	[Add result]	Not tested
Domain user signs in	Authentication succeeds	[Add result]	Not tested
GPO applies	gpresult shows expected policy	[Add result]	Not tested

7. Troubleshooting Notes

Symptom	Checks to perform	notes
Client cannot find domain	Check client DNS points to the domain controller; verify IP connectivity and domain spelling.	IP connectivity
DNS lookup fails	Check DNS service, adapter settings, and domain DNS records.	Domain DNS records
Client cannot reach server	Check both VMs use compatible VirtualBox network settings and addresses/subnets.	Virtualbox network setting
GPO does not apply	Check GPO link, OU placement, security filtering, and gpresult output.	Gpresult output

8. Security and Good Practice

- Keep the lab isolated and do not expose a test domain controller directly to the public internet.
- Use unique, strong passwords for lab accounts and store recovery credentials securely.
- Never commit passwords, recovery keys, private certificates, tokens, or personal information to GitHub.
- Use only test users and non-production data.
- Keep installation media and scripts from trusted sources; inspect scripts before running them.
- Record the actual result honestly; do not claim a task is complete until it has been tested.

9. Screenshot Checklist

- VirtualBox VM settings (avoid showing unrelated private information).
- Windows Server 2025 desktop / Server Manager.
- Server computer name and static IP configuration.
- AD DS role installation and domain configuration.
- Active Directory Users and Computers showing lab OUs and test accounts.
- DNS lookup results.
- Windows client joined to the domain.
- Group Policy configuration and gpresult output.
- Successful test results.

10. Repository Structure

Suggested GitHub layout:

```
windows-server-2025-active-directory-lab/
├── README.md
├── documentation/
│   └── Windows-Server-2025-Project-Documentation.docx
├── screenshots/
├── powershell-scripts/
└── diagrams/
    └── network-topology.png
```

11. CV Project Entry

Windows Server 2025 – Active Directory Administration Lab | GitHub: [Insert repository URL]

Configured a Windows Server 2025 virtual lab and documented server networking, AD DS, DNS, domain user management, Group Policy, and client-domain testing. Technologies: Windows Server 2025, VirtualBox, Active Directory, DNS, and PowerShell.

12. Completion Summary

- Final project status: In progress

GitHub repository URL: [Add URL]

Last updated: [Add date]